

DTSA 5707

# Deep Learning Applications for Computer Vision

|                                                |                                |               |
|------------------------------------------------|--------------------------------|---------------|
| FORMAT                                         | CATEGORY                       | CREDITS       |
| Online (DTSA)                                  | Elective                       | 1             |
| SECTION                                        | BEST ROLE MATCH                | AVG ALIGNMENT |
| Section 4 – Elective                           | Applied Scientist (72.7 / 100) | 64.5 / 100    |
| SPECIALIZATION Introduction to Computer Vision |                                |               |

64

CURRENT AVG SCORE  
across 6 job roles

94

POTENTIAL SCORE  
with recommended additions

+39

MAX SCORE GAIN  
best single-role improvement

## CURRENT COURSE DESCRIPTION

Students will learn about Computer Vision as a field of study and research. They explore several Computer Vision tasks and suggested approaches, from the classic Computer Vision perspective. They'll be introduced to Deep Learning methods and apply them to some of the same problems. They will analyze the results and discuss advantages and drawbacks of both types of methods. Examples of Computer Vision tasks where Deep Learning can be applied include: image classification, image classification with localization, object detection, object segmentation, facial recognition, and activity or pose estimation.

## INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE              | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|------------------------------|------------|----------|
| Applied Scientist         | <div><div></div></div> 72.7 | <div><div></div></div> 100.0 | +27.3      | MEDIUM   |
| AI Engineer               | <div><div></div></div> 70.9 | <div><div></div></div> 100.0 | +29.1      | MEDIUM   |
| MLOps Engineer            | <div><div></div></div> 67.9 | <div><div></div></div> 99.1  | +31.3      | HIGH     |
| Machine Learning Engineer | <div><div></div></div> 63.3 | <div><div></div></div> 80.7  | +17.4      | MEDIUM   |
| Data Scientist            | <div><div></div></div> 57.4 | <div><div></div></div> 96.2  | +38.8      | HIGH     |
| ML Engineer               | <div><div></div></div> 54.6 | <div><div></div></div> 86.3  | +31.7      | HIGH     |

## RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**System Design** 4 of 6 roles

systems

**Deep Learning** 2 of 6 roles

cnn

training

**Business Communication** 1 of 6 roles

insights

**Experiment Tracking** 1 of 6 roles

tracking

## PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Deep Learning content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.
- LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.
- LOW** **Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.

## GAP KEYWORDS BY ROLE

**Applied Scientist**

72.7 → 100.0 (+27.3 pts)

literature diffusion models technical data systems industry design

**AI Engineer**

70.9 → 100.0 (+29.1 pts)

data models build technical systems technology machine information

**MLOps Engineer**

67.9 → 99.1 (+31.3 pts)

data text systems machine tracking training tools models

**Machine Learning Engineer**

63.3 → 80.7 (+17.4 pts)

data tools design science models applications machine performance

**Data Scientist**

57.4 → 96.2 (+38.8 pts)

data product insights different build

**ML Engineer**

54.6 → 86.3 (+31.7 pts)

cnn data machine systems models world build impact

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with  $\geq 200$  applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.